

A ratio-locked radio signature of a universal shift in the electron mass

Justin Pulford^{1,*}

¹*Unaffiliated*

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A universal fractional shift ε in the electron mass would imprint fixed relative shifts across radio bands, with ratios set by atomic physics and independent of ε itself. That multi-band *pattern*—and the falsification it enables—is the content of this paper, not a competitive bound on the amplitude. Methanol absorption already constrains $|\Delta\mu/\mu|$ at the $\text{few} \times 10^{-7}$ level, some thirty-five times tighter than the best 21 cm reinterpretation quoted below; a single molecular comparison cannot, however, test whether a shift lies in m_e rather than in a coupling, nor whether every band shares it. We derive how such a shift would *propagate*: +2 for the hydrogen 21 cm hyperfine frequency, +1 for radio recombination lines, −1 for a dispersion-measure reconstruction (degenerate with the fitted column unless an independent electron column is supplied), −1 or −3 for synchrotron characteristic frequency according as the emitting population is labelled by Lorentz factor or by energy, and −2 for Faraday rotation. Only the 21 cm and Faraday rows are presently usable in a Gaussian multi-band fit; the other three are theoretical propagation channels, not simultaneous precision measurements available today. The pattern separates a varying electron mass from a varying fine-structure constant by sign structure, and the hydrogen-to-deuterium hyperfine ratio is an internal control on universality. We treat ε as a free amplitude to be fitted, not predicted, and commit to no mechanism. The construction is a theoretical template for multi-band programmes, not a new observational bound or an end-to-end survey forecast. A single band deviating from its assigned weight falsifies the pattern.

I. INTRODUCTION

Whether the fundamental constants vary in space or time has a long observational literature [1, 2]. Limits come from quasar absorption spectra [3, 4], from the cosmic microwave background [5, 6], and from big-bang nucleosynthesis, the Oklo natural reactor, and laboratory clock comparisons. Most of this work has concentrated on the fine-structure constant; the electron mass has attracted more recent attention as well, including in cosmological settings where a larger m_e near recombination can alter the sound horizon [6, 7]. Those motivations are background. This paper does not predict ε , does not commit to a mechanism for it, and does not claim to resolve any cosmological tension.

The tightest existing limits on a fractional shift in $\mu \equiv m_p/m_e$ already come from molecular spectroscopy, not from the radio continuum or from 21 cm absorption. Methanol absorption toward PKS1830–211 bounds $|\Delta\mu/\mu|$ at a $\text{few} \times 10^{-7}$ [8, 9] — some thirty-five times tighter than the best published 21 cm comparison reinterpreted as a bound on ε below. That amplitude superiority is settled; it is not what is at issue here. A single molecular comparison returns one number against one free amplitude: every hypothesis with a free ε accommodates it equally well. What a multi-band radio pattern can supply, and a single line cannot, is a *falsifiable correlation*: fixed relative shifts across independent systematics, with a sign structure that separates a varying m_e from a varying α .

Radio propagation has been used for varying constants before. Dispersion measures of localised fast radio bursts have been used to bound variation in α and in the proton-to-electron mass ratio [10], and the clustering of burst dispersion measures likewise [11]. Those analyses treat a single observable at a time. The construction used below, in which a quantity reconstructed with laboratory constants is read against something that does not share its dependence, is not new and is not confined to the radio bands. The thermal Sunyaev–Zel’dovich effect carries the Thomson cross-section, $\sigma_T \propto \alpha^2 m_e^{-2}$, so a Comptonization inferred from it inherits a weight in exactly the sense of Eq. (3). Compared against an independent X-ray determination of the same intracluster gas, that weight has been used to bound variation in α over 119 clusters [12], and over 618 for spatial rather than temporal variation [13]. What follows applies the same logic to a different set of observables, chosen so that the weights differ in sign as well as magnitude.

Different radio observables depend on the electron mass through different powers, so a universal shift in m_e does not move them together: it moves them by fixed, unequal, calculable amounts. Section III derives five such weights as a statement of how a shift would *propagate*. Only two of those rows — the 21 cm hyperfine frequency and Faraday rotation — are presently usable in a Gaussian multi-band analysis of the form Eq. (13); the recombination-line, synchrotron and dispersion-measure rows fail that presumption for reasons of kind rather than precision (Sec. VI). The five-row vector is therefore not a claim of five simultaneous measurements available today, and this paper is a theoretical template for multi-band programmes, not a new observational bound or an end-to-end survey forecast.

* pulfordj420@gmail.com

The argument is arranged as follows. Section II fixes notation and separates the two kinds of observable involved. Section III derives the five weights. Section IV shows how the pattern separates a varying electron mass from a varying fine-structure constant, and examines an isotopic control that is easy to misread. Section V states what the argument assumes. Section VI works out which rows can be entered today, the formal sensitivity of the usable pair, and where the amplitude already stands from methanol.

II. SETUP

We consider a universal, species-independent fractional shift in the electron mass,

$$m_e \longrightarrow m_e(1 + \varepsilon), \quad |\varepsilon| \ll 1, \quad (1)$$

holding all other constants fixed. We make no assumption about what produces Eq. (1) and we do not predict ε ; it is a single free amplitude. The content of this paper is the set of *relative* responses, which do not depend on it.

A. What is held fixed, and why the statement is dimensionless

A shift in a dimensionful constant is not by itself physically meaningful: the magnitude and even the sign of an inferred effect can depend on the choice of units, an objection raised explicitly against earlier 21 cm work on varying constants [14–16]. Equation (1) must therefore be read with its reference declared.

We hold the fine-structure constant and the strong scale fixed, so that Eq. (1) is a statement about the dimensionless mass ratio

$$\mu \equiv m_p/m_e, \quad \delta\mu/\mu = -\varepsilon. \quad (2)$$

Every row below then restates without reference to a unit system. The two line frequencies become dimensionless when referred to the proton mass, $\nu_{\text{hf}}/m_p \propto \alpha^4 \mu^{-2}$ and $\nu_{\text{RRL}}/m_p \propto \alpha^2 \mu^{-1}$, and are observed as apparent redshift offsets, which are dimensionless in any case. The dispersion and rotation rows are ratios of a reconstructed quantity to the true one, with the same units above and below. The synchrotron row is dimensionless when referred to the proton cyclotron frequency, $\nu_c/\nu_{c,p} \propto \mu$.

Recast this way the weight vector is unchanged, so nothing in Sec. III depends on the unit convention. We continue to write the weights against m_e because it is the more familiar form, with Eq. (2) understood throughout.

For an observable $\mathcal{O}$ we write its weight as the logarithmic derivative

$$w_{\mathcal{O}} \equiv \frac{\partial \ln \mathcal{O}}{\partial \ln m_e}, \quad \text{so} \quad \frac{\delta \mathcal{O}}{\mathcal{O}} = w_{\mathcal{O}} \varepsilon + \mathcal{O}(\varepsilon^2). \quad (3)$$

Two classes of observable respond differently, and the distinction matters for what is measurable.

Line frequencies are one class. For a single line a shift in the rest frequency is absorbed into an apparent redshift: an observer measuring ν_{obs} and assuming the laboratory ν_{rest} infers $1 + z_{\text{app}} = (1 + z)(1 + w\varepsilon)$. One line alone therefore carries no information about ε . Two lines of different weight do, since their inferred redshifts differ by a known factor.

Dispersion and rotation measures are the second class. These are not frequencies but quantities reconstructed from a measured delay or rotation angle using laboratory constants. A shift in m_e along the line of sight biases the reconstruction even when the physical column is unchanged.

The pattern derived below spans both classes, so it is not degenerate with redshift.

III. THE FIVE WEIGHTS

A. Hydrogen 21 cm hyperfine: $w = +2$

The ground-state hyperfine splitting is the Fermi contact interaction between the electron and nuclear moments [17, 18]. Its scale is the Rydberg reduced by the ratio of the nuclear to the Bohr magneton and by one further factor α^2 ,

$$\nu_{\text{hf}} \propto \alpha^2 \underbrace{\frac{\mu_N}{\mu_B}}_{\propto m_e/m_p} \underbrace{R_{\infty} c}_{\propto \alpha^2 m_e} \propto g_p \alpha^4 \frac{m_e^2}{m_p}, \quad (4)$$

giving $w = +2$.

B. Radio recombination lines: $w = +1$

Transitions between high principal quantum numbers are pure Rydberg [19], $\nu = R_{\infty} c (n_1^{-2} - n_2^{-2})$ with $R_{\infty} \propto \alpha^2 m_e$, so

$$\nu_{\text{RRL}} \propto m_e, \quad w = +1. \quad (5)$$

C. Dispersion measure: $w = -1$

The group delay of a signal traversing a cold plasma is [20]

$$t_{\text{DM}} = \frac{e^2}{2\pi m_e c} \frac{1}{\nu^2} \int n_e dl. \quad (6)$$

The measured quantity is t_{DM} ; the reported dispersion measure is obtained by dividing out [21] the *laboratory* constant $e^2/2\pi m_e^{\text{lab}} c$. If the true electron mass along the path is $m_e^{\text{lab}}(1 + \varepsilon)$, the delay produced by a fixed physical

column $\int n_e dl$ is smaller by $(1 + \varepsilon)^{-1}$, and the inferred column inherits that factor:

$$\text{DM}_{\text{inferred}} = \left(\int n_e dl \right) (1 - \varepsilon) + \mathcal{O}(\varepsilon^2), \quad w = -1. \quad (7)$$

The bias is in the *reconstruction*, not in the plasma. This row therefore tests the shift against any independent determination of the same electron column.

D. Synchrotron characteristic frequency: convention-dependent (−1 or −3)

The characteristic frequency of synchrotron emission is [22]

$$\nu_c = \frac{3}{2} \gamma^2 \frac{eB}{2\pi m_e c}. \quad (8)$$

This is the one convention-dependent entry, and the convention is not a bookkeeping detail: it changes the weight by a factor of three and must sit front and centre before the row is used. Holding the magnetic field and the emitting particle's Lorentz factor fixed gives $w = -1$ from the explicit m_e in Eq. (8). Holding the particle's energy fixed instead lets $\gamma = E/m_e c^2$ float, contributing a further -2 , so that

$$w_{\text{sync}} = \begin{cases} -1, & \text{fixed } \gamma \text{ (fixed-field reading),} \\ -3, & \text{fixed } E \text{ (fixed-energy reading).} \end{cases} \quad (9)$$

Neither value is default physics. Real source modelling often labels the emitting population by energy (injection spectra, cooling breaks, equipartition arguments), and under that labelling the row is -3 ; under a fixed- γ labelling it is -1 . The choice must be fixed by the physical situation before the row is read, and if the population cannot be labelled unambiguously the row is ill-posed as a test of ε . For bookkeeping we quote -1 in Eq. (11) and Table I under an explicit fixed- γ convention; that is a declaration, not a claim that fixed- γ is the correct description of any particular source. We return to this in Sec. V.

E. Faraday rotation: $w = -2$

The rotation measure of a magnetised plasma is [23]

$$\text{RM} = \frac{e^3}{2\pi m_e^2 c^4} \int n_e B_{\parallel} dl, \quad w = -2, \quad (10)$$

again a bias in the reconstructed quantity rather than in the medium.

TABLE I. Response of each observable to a fractional shift in the electron mass and in the fine-structure constant, $w \equiv \partial \ln \mathcal{O} / \partial \ln X$. The two columns are not proportional, so the hypotheses are distinguishable; the sign structure separates them without reference to precision.

observable	w for m_e	w for α
21 cm hyperfine	+2	+4
radio recombination line	+1	+2
dispersion measure (inferred)	−1	+1
synchrotron ν_c (fixed γ ; −3 at fixed E)	−1	+1/2
Faraday rotation measure	−2	+3/2

F. The pattern

Collecting Eqs. (4)–(8):

$$\boxed{w_{21} : w_{\text{RRL}} : w_{\text{DM}} : w_{\text{sync}} : w_{\text{RM}} = +2 : +1 : -1 : -1 : -2} \quad (11)$$

Every entry follows from a textbook scaling once its convention is fixed; the synchrotron entry is the sole convention-dependent weight, quoted here as -1 under the fixed- γ declaration of Sec. III. The amplitude ε is common to all five and cancels from every ratio, so Eq. (11) is a prediction about *correlations* — how a shift would propagate across bands — that can be tested without knowing the size of the shift.

IV. SEPARATION FROM A VARYING FINE-STRUCTURE CONSTANT

The competing hypothesis is a shift in α rather than m_e . The two are separable because they produce different weight vectors across the same five bands. Working in Gaussian units, where $e^2 = \alpha \hbar c$ and hence $e \propto \alpha^{1/2}$, the same derivations give the α -weights in Table I.

The distinction is qualitative before it is quantitative. Every α -weight is positive: a shift in the fine-structure constant moves all five observables in the same direction. The m_e -weights change sign across the set, raising two observables and lowering three. Measuring any two bands of opposite m_e -weight therefore separates the hypotheses by the sign of the correlation alone.

The reason is visible in the two reconstructed quantities. The dispersion delay of Eq. (6) carries $e^2/m_e \propto \alpha/m_e$, so α and m_e enter with opposite signs; the same holds for the rotation measure, where $e^3/m_e^2 \propto \alpha^{3/2}/m_e^2$. The line frequencies, by contrast, carry both constants with positive powers. No overall rescaling maps one column of Table I onto the other: the ratios w_{α}/w_{m_e} across the five rows are 2, 2, $-1/2$ and $-3/4$.

A. The hydrogen–deuterium control

The hyperfine frequencies of hydrogen and deuterium both take the form of Eq. (4) with their own nuclear factors. Their ratio is therefore

$$\frac{\nu_{\text{hf}}(\text{H})}{\nu_{\text{hf}}(\text{D})} = \frac{C_{\text{H}} g_p / m_p}{C_{\text{D}} g_d / m_d}, \quad (12)$$

in which both α and m_e cancel identically, the angular-momentum coefficients C depending only on the nuclear spins. Logarithmic differentiation confirms $\partial \ln(\text{ratio}) / \partial \ln m_e = \partial \ln(\text{ratio}) / \partial \ln \alpha = 0$.

The ratio is therefore not a discriminant between a varying m_e and a varying α , since both preserve it. What it tests is universality: whether whatever shifts does so identically for the two isotopes. It excludes species-dependent variation, which is a useful null in its own right, and it is easily misread as evidence for one constant over the other.

V. WHAT IS ASSUMED

The weights in Eq. (11) follow from standard atomic and plasma physics, but the test built on them carries assumptions that should be stated.

Universality. The shift is taken to act identically on every electron. A species- or environment-dependent variation would break the pattern, and the hydrogen–deuterium ratio of Sec. IV is the control for this.

Other constants fixed. We vary m_e alone. A simultaneous variation in α or in the proton mass adds its own weights to each row, and the observed pattern would then be a superposition. Because the weight vectors differ between m_e and α , a sufficiently well-measured set of bands can in principle separate the components; with few bands it cannot.

Linearity. Existing bounds on a varying electron mass from big-bang nucleosynthesis and the microwave background place $|\varepsilon|$ at the percent level or below [24, 25]. We work to first order in ε . For $|\varepsilon| \lesssim 10^{-2}$ the quadratic terms enter at the 10^{-4} level, below the precision of any of the measurements considered here.

Constancy along the path. Each observable integrates along a line of sight or over an epoch. The weights apply cleanly if ε is constant over the relevant interval. If it varies, each row picks up its own weighting of the variation and the ratios are modified in a row-dependent way. Comparisons at matched redshift are therefore preferable to comparisons across widely separated epochs.

The synchrotron convention. As noted in Sec. III, this row reads -1 or -3 according to whether the emitting population is labelled by Lorentz factor or by energy. The -1 entry in Eq. (11) is a declared fixed- γ convention, not a preferred physical default. Real source modelling may make the labelling ambiguous, in which case the row is ill-posed and should be dropped from any fit rather than entered under an unspoken choice.

VI. SENSITIVITY

Suppose each band i yields a measured fractional shift d_i with Gaussian uncertainty σ_i , against the prediction $d_i = w_i \varepsilon$. The maximum-likelihood amplitude and its error are

$$\hat{\varepsilon} = \frac{\sum_i w_i d_i / \sigma_i^2}{\sum_i w_i^2 / \sigma_i^2}, \quad \sigma_\varepsilon = \left(\sum_i \frac{w_i^2}{\sigma_i^2} \right)^{-1/2}. \quad (13)$$

The weights enter quadratically, so the high- $|w|$ rows dominate. Only two rows are presently usable in Eq. (13), for reasons of kind set out immediately below; among those, the 21 cm and Faraday rows are the extremes of the weight set and give $\sigma_\varepsilon = \sigma / \sqrt{8} \simeq 0.35 \sigma$ at common fractional precision σ . That pair, not a five-band programme, is the realistic near-term test. For orientation only, if all five bands were somehow measured at common precision one would have $\sum_i w_i^2 = 11$ and $\sigma_\varepsilon = \sigma / \sqrt{11} \simeq 0.30 \sigma$; that figure is arithmetic on the weight vector, not a survey forecast, and we do not treat it as a reachable target.

Equation (13) presumes each band supplies an independent Gaussian σ_i ; three of the five rows do not. Radio recombination frequencies are *computed* from the Rydberg formula and therefore already carry m_e , so an observed line compared against its own computed rest frequency returns a velocity rather than a constraint; that row enters only in differential comparison against a transition of different m_e dependence. Synchrotron carries a formal degeneracy between the field strength and the emitting population, broken in practice by assuming energy equipartition, so its uncertainty is dominated by a modelling choice that is itself degenerate with the quantity being weighed — the resulting normalisation correction does not have a fixed sign, and the -1 versus -3 convention of Sec. III may leave the row ill-posed for a real source. The dispersion-measure row fails the same presumption, for the reason set out below: a constant ε is absorbed exactly by the fitted dispersion measure, so the row reports the precision of an independent electron-column determination rather than a σ_i of its own.

We therefore treat only the two rows referred to a fixed laboratory rest frequency (21 cm and Faraday rotation) as presently measurable. The five-row pattern of Sec. III is unaffected: it is a statement about how a shift propagates, not a claim of five simultaneous Gaussian measurements available today. What is reduced is the number of rows that can presently be turned into a fit.

Pairs of opposite m_e -weight are preferable for a reason independent of their statistical weight. A systematic common to both bands, entering as a shared multiplicative offset, contributes equally to d_{21} and d_{RM} and therefore cancels in the difference that carries the signal, whereas it would be absorbed into $\hat{\varepsilon}$ by a same-sign pair. The sign structure of Table I is thus useful twice: it separates the m_e and α hypotheses, and it suppresses common-mode systematics.

A. Where the rows currently stand

The two presently measurable rows are quoted below with their statistical figures and, separately, the systematic that actually binds each; in both cases the systematic exceeds the statistical precision by roughly half an order of magnitude, and in each it arises from a different physical effect. The dispersion-measure row is set out between them, not as a third constraint but because its exclusion rests on a degeneracy the reader should be able to check rather than on an assertion.

The 21 cm row is constrained by comparing hyperfine absorption against optical metal lines in the same absorber. The tightest such analysis reports $\Delta x/x = -(0.1 \pm 1.3) \times 10^{-6}$ over four systems at $1.17 < z < 1.56$, where $x \equiv g_p \alpha^2 / \mu$ [26]. Since $\mu = m_p/m_e$, holding α and the proton moment fixed gives $\delta \ln x = -\delta \ln \mu = \varepsilon$, so that measurement is a measurement of the amplitude itself:

$$\varepsilon = -(0.1 \pm 1.3) \times 10^{-6} \quad (1.17 < z < 1.56), \quad (14)$$

statistical only. An earlier analysis of the same combination quotes a maximum systematic of 6.7×10^{-6} [27], arising because the radio and optical absorption need not sample the same gas along a finite-width sightline — which is why the tighter analysis restricts itself to milliarsecond-unresolved background sources. Added in quadrature the two give $\varepsilon = -(0.1 \pm 6.8) \times 10^{-6}$, or $|\varepsilon| < 1.4 \times 10^{-5}$ at 2σ . This is a reinterpretation of a published measurement under the assumptions of Sec. V, not a new one. The systematic exceeds the statistical error by a factor of five — exactly the situation a ratio test across bands with independent systematics is meant to improve on.

The dispersion-measure row has by far the best raw precision and, for the same reason as the recombination-line row, the least of it is usable. Median DM uncertainties reach $\sim 10^{-5} \text{ pc cm}^{-3}$ in low-frequency monitoring, which against typical pulsar-timing-array dispersion measures of $10\text{--}100 \text{ pc cm}^{-3}$ is a fractional precision of $10^{-7}\text{--}10^{-6}$. That precision does not transfer to ε . A pulsar timing model fits an infinite-frequency arrival time and a dispersion measure from $t(\nu) = t_\infty + K \text{ DM}/\nu^2$, and a universal shift $m_e \rightarrow m_e(1 + \varepsilon)$ rescales the delay's coefficient while leaving its ν^{-2} shape untouched. The fit therefore absorbs it exactly, returning $\text{DM}_{\text{fit}} = (\int n_e dl)/(1 + \varepsilon)$ with t_∞ unchanged and no timing residual at any frequency coverage. A constant ε is thus degenerate with the fitted dispersion measure, and improving the DM measurement improves DM_{fit} while saying nothing about the shift. The row constrains ε only in the sense already stated in Sec. III: against an independent determination of the same electron column, whose precision, not the DM precision, sets σ_ε .

That route has been taken. The fast-radio-burst analyses cited in Sec. I compare the observed dispersion measure against the value predicted from $\Omega_b h^2$ and an assumed intergalactic baryon fraction, which is exactly the

external column referred to above. Seventeen bursts with supernovae in a runaway-dilaton parametrization give $\Delta\alpha/\alpha$ at the 10^{-2} level [28]; 50 bursts over $0.004 < z < 1.02$ tighten this to $\Delta\alpha/\alpha \simeq 2 \times 10^{-5}$ [10]. One caution about reading those results as mass constraints. The quoted $\Delta\mu/\mu \simeq -1 \times 10^{-5}$ is not independent of the α result but derived from it, through an assumed relation between mass and coupling variation, so it does not constrain ε at fixed α — the case considered here. The corresponding mass-only analysis has not to our knowledge been performed, and its limiting uncertainty would be the baryon fraction rather than the dispersion measures.

What the timing literature does bound is the *variation* of ε . Mis-estimated dispersion produces infinite-frequency arrival-time offsets of order $20 \mu\text{s}$ for high-DM pulsars, correlated on month timescales [29, 30]; read as a bound on epoch-to-epoch drift this gives $\sim 2 \times 10^{-6}$ at high DM and low frequency, falling to $\sim 3 \times 10^{-4}$ at 1400 MHz and low column. We therefore do not fold the 10^{-7} figure into any sensitivity estimate, and we do not treat this row as carrying a constant- ε constraint at all.

The Faraday row is set by rotation-measure catalogue precision. The deepest current grid reports a median RM uncertainty of 0.06 rad m^{-2} across 2461 extragalactic sources, with a possible residual systematic of up to 0.3 rad m^{-2} after ionospheric correction [31].

B. Where the amplitude actually stands

The strongest existing bound on ε comes from none of these rows, and it is stated first so that the paper is not read as competing on amplitude. Methanol absorption in the $z = 0.88582$ lens toward PKS1830–211 gives $|\Delta\mu/\mu| \lesssim 4 \times 10^{-7}$ at 2σ over $0 < z \leq 0.886$ [8]. The methanol transitions carry μ alone — no α , no proton moment — so that limit constrains ε under weaker assumptions than Eq. (14) requires, and it is some thirty-five times tighter. An independent analysis of the same source at three telescopes, using ten transitions, agrees: $\Delta\mu/\mu = (-1.0 \pm 0.8_{\text{stat}} \pm 1.0_{\text{sys}}) \times 10^{-7}$ [9], which is $|\Delta\mu/\mu| < 3.6 \times 10^{-7}$ at 2σ and is likewise dominated by its systematic. Matching that figure with the 21 cm and Faraday pair would require a fractional precision of $\sigma \simeq 1.1 \times 10^{-6}$ per band; the two rows that can presently be entered do not improve the limit on the amplitude, and we do not claim that they do.

What they supply instead is something a single molecular comparison cannot. One transition returns one number against one free amplitude ε ; the fit is exact and nothing is left over, so every hypothesis with a free amplitude accommodates the result equally well. Two rows of different weight return two numbers for the same amplitude, leaving one residual degree of freedom that the hypotheses do not share — and it is that residual, not the amplitude, that Eq. (13) calculates the sensitivity of. Because the count does not involve σ , a tighter bound on

the amplitude does not diminish the pattern test. The methanol analyses already illustrate the same point that limits the 21 cm row: a tighter statistical limit from one line pair was set aside in favour of the weaker figure from three transitions whose profiles agree, on the grounds that only those demonstrably sample the same gas. Both of the best available constraints on ε are limited by sightline structure rather than by photon noise.

This paper is a theoretical template for multi-band programmes: it supplies weights, a falsification rule, and an honest account of which rows enter a Gaussian fit today. It is not a new observational bound, and it is not an end-to-end survey forecast.

VII. FALSIFICATION

The pattern is falsified by any single band deviating from its assigned weight, once two or more usable bands are measured along the same sightline or at matched redshift. With only the 21 cm and Faraday rows presently enterable, the near-term falsifier is a pair whose residual is inconsistent with a common ε ; the full five-weight vector remains the theoretical target if and when the other rows become well posed.

Two outcomes distinguish the hypothesis from the alternatives. A pattern consistent with $+2 : +1 : -1 : -1 : -2$ across the bands that can be entered supports a universal shift in m_e . A pattern with the same signs but different magnitudes is evidence for a mixed variation, or for a variation in α , and the weight vectors separate the cases. Uncorrelated shifts, or a single anomalous band with the others null, are not evidence for this hypothesis

and should be treated as instrumental or astrophysical in origin.

VIII. CONCLUSION

A universal fractional shift in the electron mass would produce a correlated signature across radio observables whose relative sizes are fixed by atomic physics. The ratios $+2 : +1 : -1 : -1 : -2$ state how such a shift would propagate; they are independent of the amplitude, so the structure can be tested without a prediction for that amplitude and without commitment to any mechanism producing it. The pattern differs from that produced by a varying fine-structure constant, and the hydrogen–deuterium hyperfine ratio provides an internal control on universality that both hypotheses pass and a species-dependent variation fails.

Only two of the five rows — 21 cm and Faraday rotation — can be entered in a Gaussian multi-band fit at present; the other three remain theoretical propagation channels. Methanol absorption already bounds the amplitude some thirty-five times more tightly than the best published 21 cm comparison ($|\varepsilon| < 1.4 \times 10^{-5}$ over $1.17 < z < 1.56$, systematic five times the statistical error). Neither figure is improved here, and this paper is not a survey forecast. The case for the pattern is not that it measures ε better, but that a limit on μ is a single number, which cannot say whether the shift lies in the electron mass or in the coupling, nor whether every band shares it. Fixed weights of two signs answer both questions; a single band off its assigned weight falsifies the pattern.

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